CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 9-2025: A 59-Year-Old Man with Hepatocellular Carcinoma

Kathleen E. Corey, M.D., M.P.H., M.M.Sc.,[1,2] David M. Dudzinski, M.D.,[1,2]
Alexander R. Guimaraes, M.D.,[3,4] and Mari Mino-Kenudson, M.D.[5,6]

Author affiliations are listed at the end of the article.

N Engl J Med 2025;392:1216-27.
DOI: 10.1056/NEJMcpc1909622

CME

PRESENTATION OF CASE

Dr. David M. Dudzinski: A 59-year-old man with a history of obesity, dyslipidemia, and hypertension was evaluated in the oncology clinic of this hospital because of hepatocellular carcinoma.

Approximately 12.5 years before the patient's evaluation at this hospital, elevated blood levels of aspartate aminotransferase and alanine aminotransferase were incidentally noted by his primary care physician at another hospital (Table 1). The blood levels of electrolytes and thyrotropin were normal, as were the results of renal-function tests. Because the patient had reported no history of alcohol use, a presumptive diagnosis of metabolic dysfunction–associated steatotic liver disease (MASLD, previously known as nonalcoholic fatty liver disease) was made.

Nearly 3 years later, the patient was evaluated at a second hospital because of malaise, fatigue, and dark-red stools. Esophagogastroduodenoscopy showed changes consistent with bile reflux gastritis; there was no evidence of varices or portal hypertensive gastropathy. Colonoscopy showed no abnormalities except for a small polyp in the transverse colon, which was removed. A diagnosis of *Helicobacter pylori* infection was made, and treatment with antibiotic agents and omeprazole was started. The blood levels of aspartate aminotransferase and alanine aminotransferase remained elevated, and additional testing was performed.

Testing for hepatitis B surface antigen and core antibodies was negative, as was testing for antibodies to hepatitis C virus. Testing for hepatitis B surface antibodies was positive. The blood globulin level was normal; other laboratory test results are shown in Table 1. Abdominal ultrasonography reportedly showed a normal-sized liver with heterogeneous parenchyma, no dilated intrahepatic or extrahepatic ducts, splenomegaly (with the spleen measuring 16.7 cm in the greatest dimension; reference value, <13.2), and prominent vessels in the splenic hilum that were consistent with varices. The patient was referred for evaluation in a gastroenterology clinic.

Six months later, the patient was evaluated in the gastroenterology clinic of a third hospital. He again reported no history of alcohol use. The only medication he had been taking was irbesartan, and there was no history of treatment with glucocorticoids. The blood pressure was 152/82 mm Hg. The weight was 115 kg, the height 175 cm, and the body-mass index (the weight in kilograms divided by the square of the height in meters) 37.6. The sclerae and skin were anicteric. The abdomen was obese, the liver extended across the right upper quadrant past the midline, and the spleen was not palpable. Spider angiomas were present on the chest; the remainder of the examination was normal. Laboratory test results are shown in Table 1.

Additional diagnostic testing was performed.

DISCUSSION OF INITIAL DIAGNOSIS

Dr. Kathleen E. Corey: I participated in the care of this patient and am aware of the final diagnosis. At 47 years of age, this man with obesity, dyslipidemia, and hypertension, but with no history of alcohol use, was found on routine testing to have elevated levels of aspartate aminotransferase and alanine aminotransferase and was given a presumptive diagnosis of MASLD. More than 3 years later, a physical examination showed spider angiomas on the chest and hepatomegaly, findings consistent with chronic liver disease. Six months earlier, a normal blood albumin level and a normal international normalized ratio had suggested intact hepatic synthetic function. The patient had thrombocytopenia, which can be seen with portal hypertension and associated splenomegaly, resulting from splenic sequestration of platelets as well as impaired production of platelets due to decreased thrombopoietin production. Imaging showed heterogeneous echotexture of the liver, which is suggestive of fatty liver infiltration or chronic liver disease, and also showed splenomegaly and splenic varices, which are consistent with portal hypertension and may indicate cirrhosis.

Patients with incidentally noted abnormal hepatic aminotransferase levels or with evidence of fatty liver infiltration on imaging — a finding indicative of steatotic liver disease — are often referred to hepatology clinics for evaluation of the cause and management of the condition. Steatotic liver disease encompasses various causes of steatosis, including MASLD, alcohol-associated liver disease (ALD), and metabolic and alcohol–associated liver disease (metALD).

STEATOTIC LIVER DISEASE

When evaluating a patient for suspected steatotic liver disease, it is important to first rule out hepatitis C virus infection and medication-induced MASLD, both of which can cause fatty infiltration of the liver. Next, it is important to identify the type of steatotic liver disease by assessing for cardiometabolic risk factors and quantifying weekly alcohol consumption. MASLD is defined by the presence of steatosis and at least one cardiometabolic risk factor, in the absence of excess alcohol use. ALD is defined by the presence of steatosis and excess alcohol use (>140 g per week for women or >210 g per week for men), in the absence of cardiometabolic risk factors. MetALD is diagnosed when the patient has steatosis, at least one cardiometabolic risk factor, and excess alcohol use.

Testing for antibodies to hepatitis C virus was negative in this patient. He had not taken any medications associated with medication-induced MASLD, such as a long-term course of glucocorticoids, methotrexate, amiodarone, or tamoxifen. He consistently reported no alcohol consumption, which ruled out the possibility of ALD and metALD.

METABOLIC DYSFUNCTION–ASSOCIATED STEATOTIC LIVER DISEASE

The patient had several cardiometabolic risk factors that would meet the criteria for MASLD as well as metabolic dysfunction–associated steatohepatitis (MASH, previously known as nonalcoholic steatohepatitis), the progressive form of MASLD. His risk factors included obesity, hypertension, and hypertriglyceridemia. In the absence of other causes of liver disease and in the presence of fatty infiltration of the liver on imaging, these cardiometabolic risk factors confirm that the diagnosis of MASLD was correctly made in this patient. If I had been treating the patient at the time of this diagnosis, I would have performed additional testing for MASLD-associated coexisting conditions, including a fasting lipid panel to evaluate for dyslipidemia and measurement of glucose, glycated hemoglobin, and insulin levels to evaluate for insulin resistance. Genetic disorders such as hypobetalipoproteinemia (characterized by low low-density lipoprotein cholesterol and apolipoprotein B levels) and lysosomal acid lipase deficiency might also be considered.

Table 1. Laboratory Data.*

Variable	First Hospital		Second Hospital		Third Hospital			This Hospital	
	Reference Range, Adults	12.5 Yr Earlier	Reference Range, Adults	9.75 Yr Earlier	Reference Range, Adults	9.25 Yr Earlier	1 Yr Earlier	Reference Range, Adults†	On Evaluation
Aspartate aminotransferase (U/liter)	5–40	51	4–37	65	11–40	48	55	10–40	60
Alanine aminotransferase (U/liter)	5–40	65	4–40	63	7–40	46	48	10–55	46
Alkaline phosphatase (U/liter)	39–117	109	39–117	138	30–115	118	95	45–115	136
Total bilirubin (mg/dl)	0.1–1.2	0.9	0.2–1.0	0.6	0.2–1.3	1.1	1.8	0.0–1.0	2.3
Direct bilirubin (mg/dl)	—	—	0.0–0.3	0.3	<0.3	0.3	0.5	0.0–0.3	—
Albumin (g/dl)	—	—	3.2–5.2	3.9	—	—	—	3.5–5.2	3.7
Prothrombin time (sec)	—	—	10.0–13.0	12.9	—	—	—	10.3–13.2	15.5
Prothrombin-time international normalized ratio	—	—	NA	1.2	<1.3	1.1	1.1	NA	1.4
White-cell count (per μl)	—	—	4800–10,800	2300	4400–11,300	3700	—	4500–11,000	4700
Platelet count (per μl)	—	—	130,000–400,000	80,000	150,000–450,000	85,000	—	150,000–400,000	67,000
Iron (μg/dl)	—	—	45–160	75	50–160	152	—	45–160	220
Total iron-binding capacity (μg/dl)	—	—	228–428	378	250–430	388	—	228–428	298
Ferritin (ng/ml)	—	—	—	—	41–306	421	—	30–300	558
Antinuclear antibody	—	—	<1:40, negative	<1:40	Negative	Negative	—	Negative at 1:40 and 1:160	Negative at 1:40 and 1:160
Smooth-muscle antibody	—	—	<1:20, negative	<1:20	—	—	—	Negative at 1:20	Positive at 1:40
Perinuclear ANCA	—	—	<1:20, negative	<1:20	—	—	—	—	—
Cytoplasmic ANCA	—	—	<1:20, negative	<1:20	—	—	—	—	—
IgG (mg/dl)	—	—	694–1618	1287	—	—	—	614–1295	1469
IgA (mg/dl)	—	—	81–463	385	—	—	—	69–309	518

IgM (mg/dl)	—	48–271	304	—	—	—	53–334	329
Serum protein electrophoresis	—	—	Normal pattern	—	—	—	—	Normal pattern
Alpha-fetoprotein (ng/ml)	—	—	—	0–10	—	11	<6.1	8.0‡
Antimitochondrial antibody	—	—	—	—	—	—	Negative at 1:20	Negative at 1:20

* To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for iron and total iron-binding capacity to micromoles per liter, multiply by 0.1791. ANCA denotes antineutrophil cytoplasmic antibody, and NA not available.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

‡ This value was obtained with an assay performed on the Roche E170 platform. Values obtained with different assay methods or kits cannot be used interchangeably.

HEPATIC FIBROSIS

Once a diagnosis of MASLD has been made, the next step is to evaluate the patient for hepatic fibrosis. Fibrosis is staged on a scale ranging from 0 to 4; stage 0 indicates the absence of fibrosis and higher stages indicate the presence of fibrosis at increasing severity, with stage 4 indicating cirrhosis. The presence of fibrosis of any stage is associated with an increased risk of death or a need for liver transplantation.[1] The first step in fibrosis staging is a noninvasive assessment of liver scarring performed with the Fibrosis-4 index. Patients with an elevated Fibrosis-4 score undergo further evaluation with elastography or enhanced liver fibrosis (ELF) testing; if neither of these tests is available, a liver biopsy is performed.

Dr. Dudzinski: Elastography and ELF testing were not available in this case. The patient was referred for a liver biopsy.

Dr. Mari Mino-Kenudson: Two months after the referral and 9 years before evaluation at this hospital, a liver biopsy performed at the third hospital revealed features that were consistent with chronic steatohepatitis (Fig. 1). There was prominent portal inflammation with bridging fibrosis (stage 3). Multiple hepatocytes contained intracytoplasmic macrovesicular fat in a patchy distribution, which is consistent with mild steatosis, and some of these hepatocytes had surrounding lymphocytes (lymphocytic satellitosis). Some hepatocytes had enlarged, irregular, clear cytoplasm that was consistent with ballooning degeneration, and some ballooned hepatocytes contained ill-formed Mallory–Denk bodies (intracytoplasmic cytoskeletal aggregates). However, the well-formed, eosinophilic, ropy Mallory bodies that are characteristic of ALD were absent. The constellation of features was consistent with steatohepatitis. Although moderate inflammation was present in a few areas, it consisted of mixed inflammatory cells without plasma-cell prominence. Trichrome staining that highlights collagen fibers showed bandlike fibrosis and advanced sinusoidal fibrosis that dissected hepatic lobules; the well-formed, regenerative hepatocyte nodules that are suggestive of established cirrhosis were not seen.

Dr. Corey: The presence of macrovesicular steatosis, hepatocyte ballooning, and lobular inflammation on liver biopsy confirmed the diagnosis of MASH. These features of hepatic injury distinguish MASH from MASLD, which is not associated

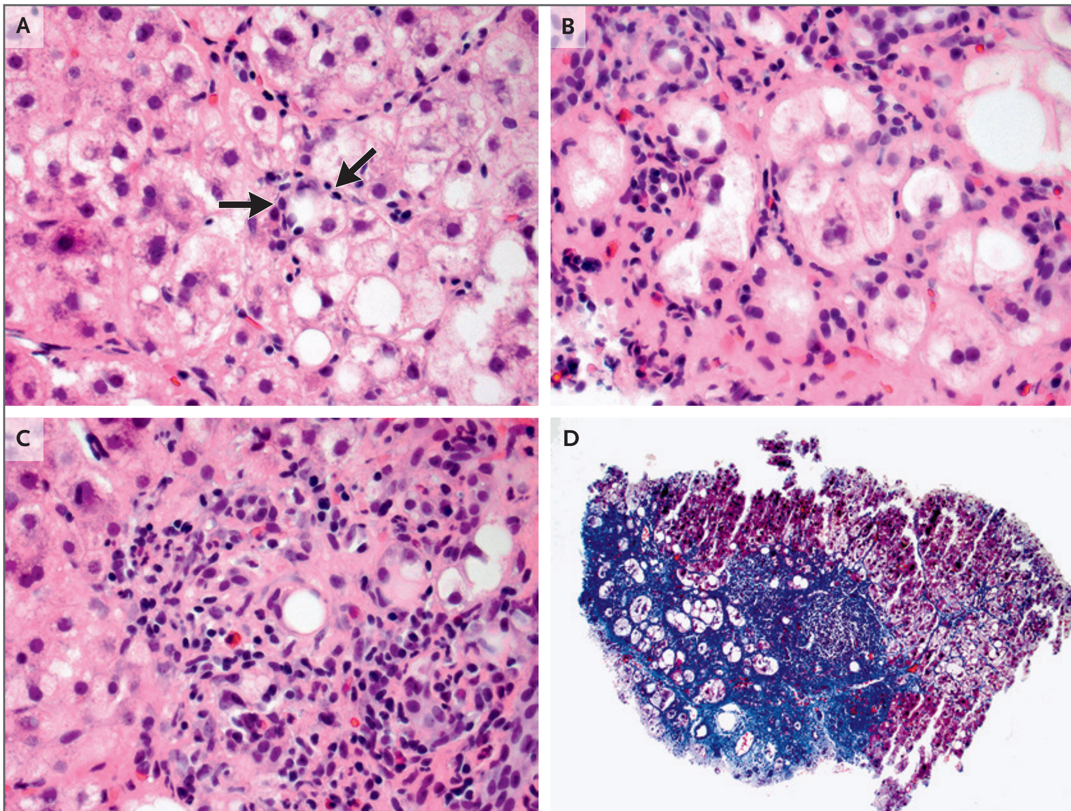

Figure 1. Liver-Biopsy Specimen Obtained 9 Years before the Current Evaluation.

Hematoxylin and eosin staining shows multiple hepatocytes with intracytoplasmic macrovesicular fat in a patchy distribution, which is consistent with mild steatosis, and also shows scattered foci of lymphocytic satellitosis (Panel A, arrows). Hepatocytes with enlarged, irregular, clear cytoplasm that is consistent with ballooning degeneration are also present in some areas (Panel B). Some of the ballooned cells contain ill-formed Mallory–Denk bodies; the well-formed, eosinophilic, ropy Mallory bodies that are characteristic of alcohol-associated liver disease are absent. Moderate inflammation that is present in a few portal tracts consists of mixed inflammatory cells without plasma-cell prominence (Panel C). Trichrome staining shows bandlike fibrosis and advanced sinusoidal fibrosis that dissects hepatic lobules; the well-formed, regenerative hepatocyte nodules that are suggestive of established cirrhosis are absent (Panel D).

with these features. Portal inflammation can also occur with MASH but is not required for the diagnosis. The most important feature noted in this patient was advanced fibrosis. Stage 3 or 4 fibrosis is associated with a risk of death or a need for liver transplantation of 1 in 3.97 to 11.97 cases and warrants aggressive management.[1]

The foundation of managing MASLD or MASH with fibrosis of any stage is weight reduction, and this patient was advised to lose weight. Loss of at least 10% of total body weight over a period of 1 year results in resolution of MASH in 90% of affected patients, as well as in reduction of fibrosis by at least one stage in 45%.[2] Resistance training, aerobic exercise, and the Mediterranean diet are associated with improvements in radiographic features of MASLD and are recommended as standard treatment. Although it was not available at the time that this patient was treated, resmetirom is a thyroid hormone receptor beta–selective agonist that is now approved for the treatment of MASH with stage 2 or 3 fibrosis.[3]

HEPATOCELLULAR CARCINOMA

Hepatocellular carcinoma is a known complication of MASH cirrhosis and can also complicate noncirrhotic MASH in rare cases. Among patients with MASH cirrhosis, the yearly cumulative incidence of hepatocellular carcinoma is 2.6%, and the incidence is increasing,[4,5] with 20,000 new cases diagnosed annually in the United States.[6] Given the risk of hepatocellular carcinoma, screening is indicated in patients with MASH cirrhosis, with imaging typically performed every 6 months.[6,7] In patients with MASH who do not have advanced fibrosis, screening for hepatocellular carcinoma has not been studied and is not routinely recommended. This patient had evidence of stage 3 fibrosis on biopsy, but the presence of varices and splenomegaly on ultrasonography suggested that he most likely had MASH cirrhosis. Therefore, screening for hepatocellular carcinoma was appropriately performed in this patient.

ADDITIONAL CASE PRESENTATION

Dr. Dudzinski: Surveillance abdominal ultrasonography for hepatocellular carcinoma was performed at the third hospital at 12-month intervals. During the 3 years after the diagnosis of MASH cirrhosis, abdominal ultrasonography did not detect any liver masses. Six years before evaluation at this hospital, the blood alpha-fetoprotein level was 12 ng per milliliter (reference range, 0 to 10), and the interval between ultrasound studies was shortened to 6 months.

Dr. Alexander R. Guimaraes: During the next 5 years, surveillance abdominal ultrasonography showed no changes. The patient had heterogeneous, mildly hyperechoic liver parenchyma, which can be seen with steatosis as well as with cirrhosis; there were no liver masses. However, 12 months before evaluation at this hospital, ultrasonography revealed a new echogenic lesion, measuring 1.2 cm in diameter, with a small hypoechoic halo that was located in liver segment 7 (Fig. 2A). Persistent findings included splenomegaly and prominent vessels in the hilum that were consistent with varices. New findings that were suggestive of progression to cirrhosis included retraction of the right hepatic lobe and enlargement of the caudate lobe.

Dr. Corey: Patients who are found to have signs of possible hepatocellular carcinoma on an ultrasound study should undergo additional imaging with either four-phase multidetector computed tomography (CT) or contrast-enhanced magnetic resonance imaging (MRI). When a lesion detected with one of these imaging techniques meets the criteria for hepatocellular carcinoma, confirmation of the diagnosis with biopsy is not needed.[6,8]

Dr. Guimaraes: One month later and 11 months before evaluation at this hospital, MRI of the liver performed at the third hospital showed features that were compatible with mild fatty liver infiltration and cirrhosis (Fig. 2B and 2C). T2-weighted imaging showed a mildly hyperintense lesion, measuring 1.2 cm in diameter (Fig. 2D). The lesion had heterogeneous arterial enhancement without rapid washout. Dynamic imaging performed with the administration of contrast material can be used to evaluate the enhancement of a lesion relative to the surrounding liver. Malignant tumors are proangiogenic and show enhancement earlier in the arterial phase than the underlying liver (early arterial enhancement) and also show washout more rapidly than the underlying liver (rapid washout). This patient did not have these imaging findings, which would be characteristic of hepatocellular carcinoma.

Five months later and 6 months before evaluation at this hospital, repeat MRI was performed without the administration of contrast material. T2-weighted imaging showed that the hyperintense lesion had increased in size, measuring 2.4 cm in diameter, with similar MRI characteristics.

Dr. Corey: This progressively enlarging T2-hyperintense lesion in liver segment 7 had imaging features that were atypical for hepatocellular carcinoma, without rapid washout. However, given the known cirrhosis and lesion growth, a biopsy was indicated.

Dr. Guimaraes: An attempted biopsy of the lesion performed under CT guidance at the third hospital was reportedly nondiagnostic. Pathological evaluation showed features that were consistent with cirrhosis and mild steatosis; no evidence of carcinoma was identified.

Three months later and 3 months before evaluation at this hospital, repeat MRI of the liver was performed. T2-weighted imaging showed that the hyperintense lesion had further increased in size, measuring 2.8 cm in diameter (Fig. 2E). Given the continued lesion growth, hepatocellular carcinoma was suspected. The blood alpha-fetoprotein level was 11 ng per milliliter. The

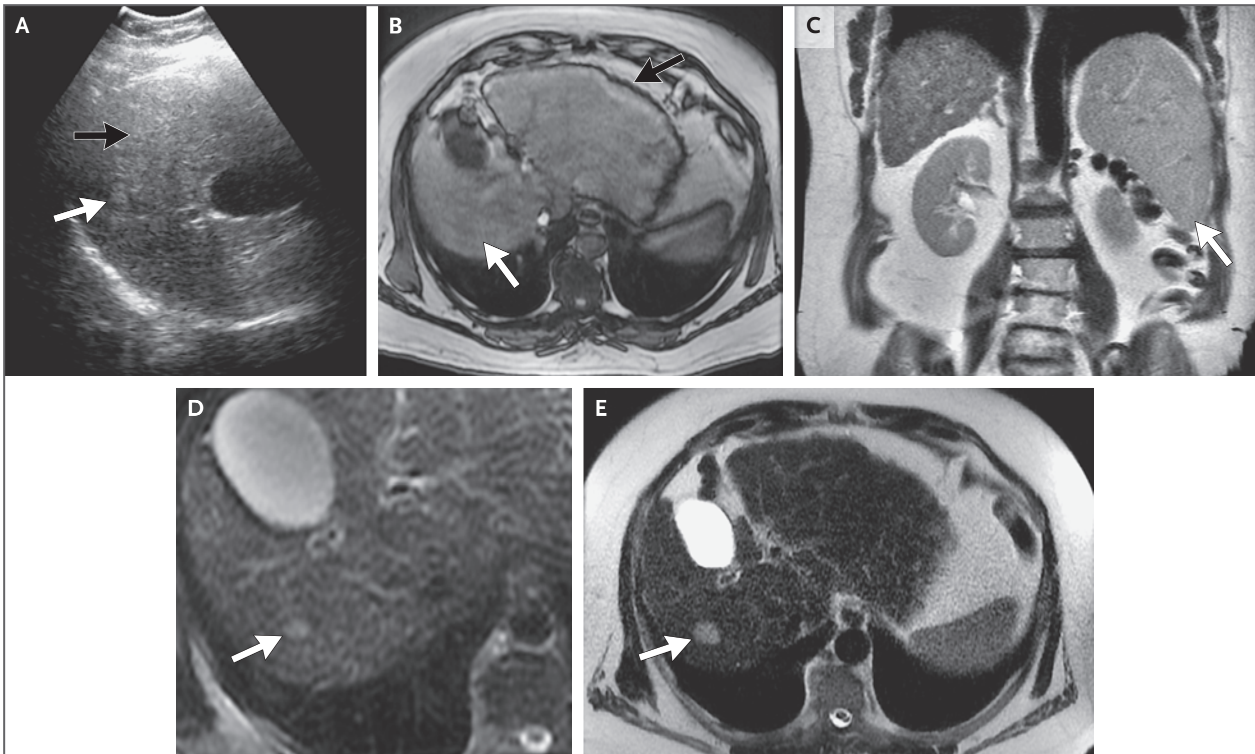

Figure 2. Ultrasound and MRI Images of the Liver.

Surveillance abdominal ultrasonography was performed 1 year before the current evaluation. An ultrasound image (Panel A) shows heterogeneous liver parenchyma (black arrow), a feature compatible with fatty liver infiltration and possibly cirrhosis, and also shows an echogenic lesion (white arrow), measuring 1.2 cm in diameter, with a hypoechoic halo that is located in liver segment 7. MRI was performed 1 month later. An axial T1-weighted image (Panel B) shows mild signal dropout at the level of the right hepatic lobe (white arrow), which is compatible with fatty liver infiltration, as well as a nodular contour of the liver parenchyma (black arrow), which is compatible with cirrhosis. A coronal T2-weighted image (Panel C) shows splenomegaly (arrow), a feature compatible with cirrhosis and portal hypertension. An axial T2-weighted image (Panel D) shows a mildly hyperintense lesion (arrow), measuring 1.2 cm in diameter, that corresponds to the abnormality seen on the ultrasound image. Repeat MRI was performed 8 months after the initial MRI study. An axial T2-weighted image (Panel E) shows that the hyperintense lesion (arrow) has increased in size, measuring 2.8 cm in diameter.

next month, a biopsy was performed under MRI guidance, and transcatheter alcohol ablation of the lesion was also performed at the time of the biopsy.

Dr. Mino-Kenudson: The biopsy specimen (Fig. 3) showed small nodules of hepatocytes separated by fibrous tissue in some areas, a finding consistent with micronodular cirrhosis. Although mild steatosis was seen, the lobular inflammation and hepatocyte ballooning that had been prominent in a previous biopsy specimen were not evident. In addition, several fragments showed a trabecular or acinar growth pattern (or both) of atypical hepatocytes. As compared with normal hepatocytes, the atypical cells were small but had a high nuclear–cytoplasmic ratio and had occasional mitoses that were indicative of malignant cells. The presence of vessels in the lesion that were not accompanied by bile ducts also indicated a neoplastic proliferation. The overall findings were consistent with hepatocellular carcinoma arising in a background of micronodular cirrhosis with mild steatosis.

Dr. Dudzinski: The patient was referred to the cancer center of this hospital. He felt well and reported no new abdominal distention, gastrointestinal bleeding, jaundice, bruising, or confusion. Other medical history included nephrolithiasis. Medications included irbesartan and propranolol. Lisinopril had caused cough; he had no other known allergies. He was married, had two adult children, and was employed in business administration. He had smoked cigarettes for 5 years but had stopped smoking 30 years earlier; he did

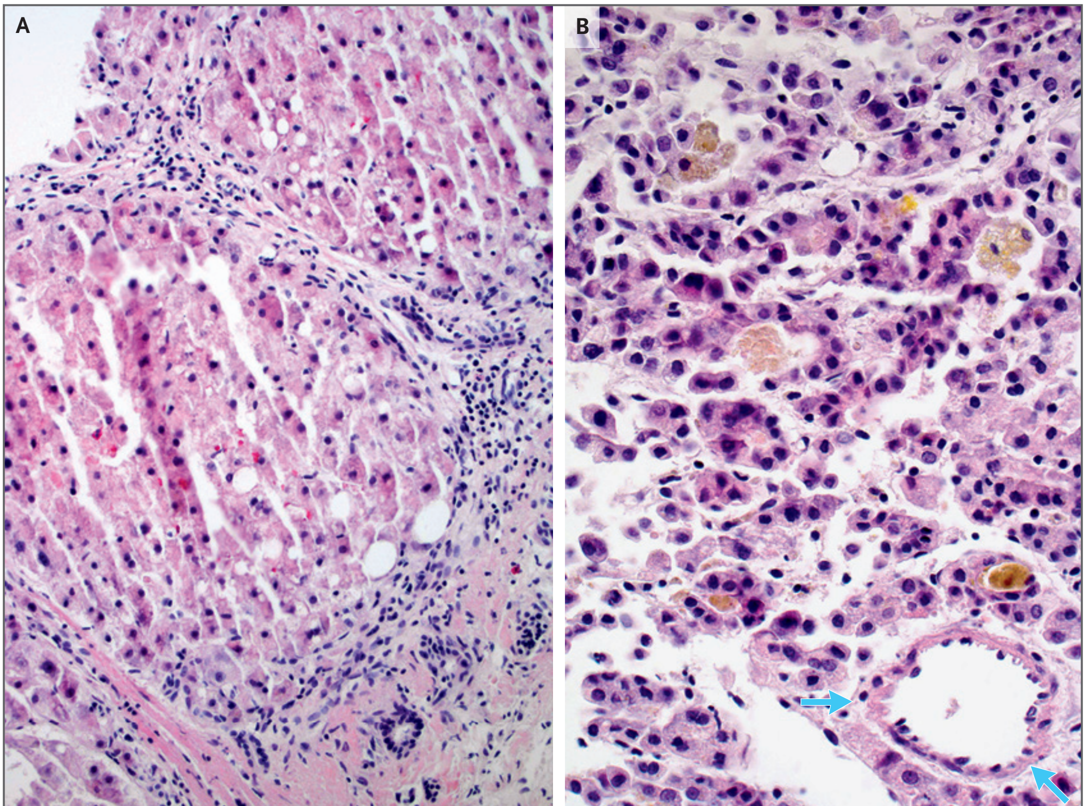

Figure 3. Liver-Biopsy Specimen Obtained 2 Months before the Current Evaluation.

Hematoxylin and eosin staining shows small nodules of hepatocytes separated by fibrous tissue in some areas, a finding consistent with micronodular cirrhosis (Panel A). Mild steatosis is present, but lobular inflammation and hepatocyte ballooning are not evident. Several fragments show a trabecular or acinar growth pattern (or both) of hepatocytes that have a high nuclear–cytoplasmic ratio and bile within acinar structures (Panel B). The presence of vessels (arrows) in the lesion that are not accompanied by bile ducts is indicative of a neoplastic proliferation.

not use illicit drugs. His father had received a diagnosis of primary sclerosing cholangitis and had died at 80 years of age, and an uncle had had hepatic cancer; his mother had hypertension, hypercholesterolemia, and coronary artery disease; and one brother had diabetes mellitus. His other siblings and his children were healthy.

On examination, the blood pressure was 138/86 mm Hg, and the heart rate was 60 beats per minute. The body-mass index was 36.4. Spider angiomas were present on the face; the conjunctivae were anicteric. The abdomen was obese, and palmar erythema and 1+ pedal edema were noted. The remainder of the examination was normal. The Model for End-Stage Liver Disease (MELD) 3.0 score was 14; scores range from 6 to 40, with higher scores indicating greater hepatic dysfunction and a higher risk of death.

Management decisions were made.

DISCUSSION OF MANAGEMENT

Dr. Corey: The patient was referred to this hospital for management of hepatocellular carcinoma complicating MASH cirrhosis. Management options for hepatocellular carcinoma include locoregional therapies such as ablation, external-beam radiation therapy, and transarterial embolization, in addition to surgical resection and liver transplantation. Surgical resection is a reasonable option for patients who do not have cirrhosis or for those who have cirrhosis without clinically significant portal hypertension. For patients who have more advanced liver disease (such as this patient with MASH cirrhosis), liver transplantation is the primary treatment option because it has the potential to cure both hepatocellular carcinoma and the underlying liver disease.

Liver transplantation performed for the treatment of hepatocellular carcinoma is increasingly common in the United States and is associated with overall positive outcomes. To be eligible for this treatment, a patient must meet the Milan criteria: the presence of either a single lesion measuring less than 5 cm in diameter or up to three lesions each measuring less than 3 cm in diameter.[9] Patients who are listed for transplantation undergo bridging therapy with locoregional therapy. Among patients who meet the Milan criteria, transplantation is associated with overall survival of 85% and recurrence-free survival of 92% at 4 years.[9]

Patients with more advanced hepatocellular carcinoma (tumors larger than those specified in the Milan criteria) have a high risk of recurrent disease after transplantation and are therefore ineligible for the treatment. Select patients with more advanced hepatocellular carcinoma that is confined to the liver may be eligible for transplantation if initial treatment results in reclassification of the tumor to a more favorable stage (downstaging). For such patients, initial treatment may consist of locoregional therapies such as ablation, external-beam radiation therapy, or transarterial chemoembolization, as well as systemic therapies such as immunotherapy with or without the use of vascular endothelial growth factor inhibitors.

For this patient with cirrhosis and newly diagnosed hepatocellular carcinoma who met the Milan criteria, liver transplantation was the treatment of choice. In New England, the expected wait time for a liver from a deceased donor to become available can often be longer than 1 year. Until he could undergo transplantation, the patient was to be monitored with assessment of the MELD 3.0 score and surveillance imaging.

Dr. Guimaraes: One week after the patient had an initial consultation at this hospital, MRI of the liver was performed after the administration of intravenous contrast material. T2-weighted imaging showed a segment 7 lesion, measuring 1.5 cm by 1.3 cm, with early arterial enhancement and hyperintensity. An adjacent area of nonenhancement was thought to possibly reflect changes that had occurred after treatment with alcohol ablation. Findings in the hepatic parenchyma were consistent with cirrhosis and fatty liver infiltration. Splenomegaly was noted, and prominent perigastric and perisplenic collateral vessels were present, along with patent portal and splenic veins. CT of the chest performed as part of the staging evaluation showed nonspecific nodules, measuring 3 mm or less in diameter.

Three months later, repeat MRI of the liver was performed after the administration of intravenous contrast material. T2-weighted imaging showed that the segment 7 lesion had increased in size, measuring 1.8 cm by 1.6 cm, with persistent early arterial enhancement.

Dr. Corey: The reduction in the size of the lesion (from 2.8 cm to 1.5 cm in diameter) after treatment with alcohol ablation suggested an initial response to therapy. However, subsequent enlargement of the lesion suggested recurrent hepatocellular carcinoma at the site of previous treatment. Until the patient could undergo liver transplantation, additional therapeutic intervention was indicated.

Dr. Guimaraes: One month later, percutaneous radiofrequency ablation was performed under CT guidance. MRI performed 1 month after treatment showed evidence of successful ablation in segment 7. However, a new lesion, measuring 1.5 cm in diameter, with early arterial enhancement and rapid washout was noted in segment 5; this finding suggested another locus of hepatocellular carcinoma.

Five weeks later, the segment 5 lesion was treated with radiofrequency ablation. MRI performed 1 month after treatment showed evidence of successful ablations in segments 5 and 7. MRI performed 3 months later showed no change in the appearance of the radiofrequency ablation zones in segments 5 and 7, but there was a new subcentimeter focus with early arterial enhancement. Chest CT performed at that time showed that the pulmonary nodules had not changed.

Dr. Dudzinski: Nineteen months after presentation to this hospital, the patient underwent successful transplantation of a liver from a deceased donor.

PATHOLOGICAL DISCUSSION

Dr. Mino-Kenudson: On pathological examination of the explanted liver (Fig. 4), the background

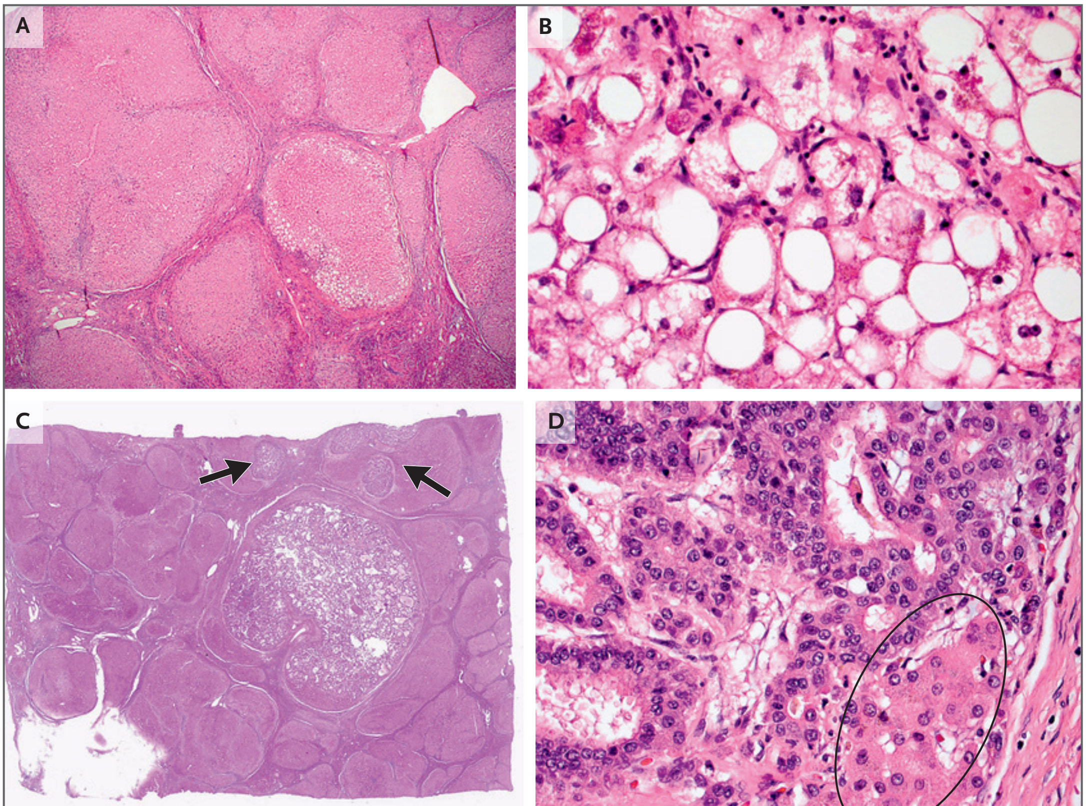

Figure 4. Liver Explant Specimen.

On hematoxylin and eosin staining, the background liver shows predominantly micronodular cirrhosis (Panel A). Rare nodules have steatosis and steatohepatitis that is characterized by ballooning degeneration of hepatocytes and lymphocytic satellitosis (Panel B). Three lesions are present in the right lobe of the explanted liver. Two of the lesions are necrotic owing to radiofrequency ablation, whereas the third has evidence of viable hepatocellular carcinoma, with a dominant mass and satellite nodules (Panel C, arrows). The mass is characterized by an acinar or trabecular growth pattern (or both) of tumor cells that have a much higher nuclear–cytoplasmic ratio than that seen in normal hepatocytes (Panel D, circled). Bile is also present within the lesion.

parenchyma showed predominantly micronodular cirrhosis. Although most of the cirrhotic nodules were devoid of fat, rare nodules had steatosis and steatohepatitis that was characterized by ballooning degeneration of hepatocytes. Three lesions were present in the right lobe of the explanted liver. Two of the lesions were completely necrotic owing to radiofrequency ablation. The third lesion had evidence of viable hepatocellular carcinoma, with a dominant mass and satellite nodules that were characterized by an acinar or trabecular growth pattern (or both) of tumor cells that had a much higher nuclear–cytoplasmic ratio than that seen in normal hepatocytes. One of the satellite nodules was surrounded in part by a muscular wall, which was suggestive of vascular invasion followed by expansile growth of the tumor in a portal tract. These findings are consistent with hepatocellular carcinoma arising in a background of predominantly micronodular cirrhosis occurring with steatohepatitis. On the basis of the presence of vascular invasion, the tumor was determined to be pathological stage 2.

ADDITIONAL MANAGEMENT AND FOLLOW-UP

Dr. Corey: Fourteen years after transplantation, the patient is doing well, without clinical or radiographic evidence of a recurrence of either cirrhosis or hepatocellular carcinoma.

Dr. Dudzinski: The cardiometabolic consequences of MASLD and MASH include insulin resistance, inflammation resulting from dysregulation of adipokine pathways, coagulopathy and effects on endothelial function, gut microflora dysbiosis, and hyperhomocysteinemia and oxidative stress. These consequences have important implications for the health of several nonhepatic organ systems, with disease risks including dyslipidemia, diabetes, hypertension, chronic kidney disease, and atherosclerotic cardiovascular and cerebrovascular disease.[10-13] MASLD and MASH are independent risk factors for cardiovascular disease, which is a leading cause of death among patients with either condition. For that reason, it is important for physicians to have an understanding of the multisystem effects of steatotic liver disease and their implications for primary care.

In the years after liver transplantation, this patient had triglyceride levels between 300 and 400 mg per deciliter (reference value, <250), with a peak level of 1080 mg per deciliter noted in the sixth year. The increase in triglyceride levels was thought to be, in part, associated with the use of sirolimus. Immunosuppressive therapy was transitioned to tacrolimus, treatment with hydrochlorothiazide was stopped, and statin therapy was started. The patient continues to receive cardiology and endocrinology care for the management of hypertension, dyslipidemia, and post-transplantation diabetes.

Patients with MASLD or MASH may also have an increased ventricular wall thickness, diastolic dysfunction, and heart failure with preserved ejection fraction.[14] Additional associated conditions include valvular sclerosis, changes in myocardial energy metabolism, an increased predilection for atrial and ventricular arrhythmia, prolongation of the corrected QT interval, and conduction defects.[11-13,15] This patient had progressive bradycardia and first-degree atrioventricular delay until 10 years after transplantation, when he presented with new atrial fibrillation with slow ventricular response. Heart rhythm monitoring showed frequent, nonsustained runs of ventricular tachycardia. Because he was unable to receive the medications necessary for rate control, a dual-chamber pacemaker was implanted.

A physician: What is the incidence of recurrent MASH after transplantation?

Dr. Corey: Recurrent MASH has been reported within 10 years after transplantation in nearly 33% of liver-transplant recipients in whom MASH was the primary disease, and recurrent cirrhosis has been reported in 3.4 to 10%.[16,17] The use of immunosuppressants such as prednisone and calcineurin inhibitors can lead to dyslipidemia, hypertension, and diabetes mellitus, which may contribute to the recurrence of MASH.

FINAL DIAGNOSIS

Metabolic dysfunction–associated steatohepatitis (MASH) cirrhosis complicated by hepatocellular carcinoma.

This case was presented at the Harvard Medical School postgraduate course “Primary Care Internal Medicine Principles and Practice,” directed by John D. Goodson, M.D.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

AUTHOR INFORMATION

[1] Department of Medicine, Massachusetts General Hospital, Boston; [2] Department of Medicine, Harvard Medical School, Boston; [3] Department of Radiology, Massachusetts General Hospital, Boston; [4] Department of Radiology, Harvard Medical School, Boston; [5] Department of Pathology, Massachusetts General Hospital, Boston; [6] Department of Pathology, Harvard Medical School, Boston.

REFERENCES

1. Angulo P, Kleiner DE, Dam-Larsen S, et al. Liver fibrosis, but no other histologic features, is associated with long-term outcomes of patients with nonalcoholic fatty liver disease. Gastroenterology 2015;149(2):389-397.e10.
2. Vilar-Gomez E, Martinez-Perez Y, Calzadilla-Bertot L, et al. Weight loss through lifestyle modification significantly reduces features of nonalcoholic steatohepatitis. Gastroenterology 2015;149(2):367-378.e5.
3. Harrison SA, Bedossa P, Guy CD, et al. A phase 3, randomized, controlled trial of resmetirom in NASH with liver fibrosis. N Engl J Med 2024;390:497-509.
4. Ascha MS, Hanouneh IA, Lopez R, Tamimi TA-R, Feldstein AF, Zein NN. The incidence and risk factors of hepatocellular carcinoma in patients with nonalcoholic steatohepatitis. Hepatology 2010;51:1972-8.
5. Baffy G, Brunt EM, Caldwell SH. Hepatocellular carcinoma in non-alcoholic fatty liver disease: an emerging menace. J Hepatol 2012;56:1384-91.
6. El-Serag HB. Hepatocellular carcinoma. N Engl J Med 2011;365:1118-27.
7. Chalasani N, Younossi Z, Lavine JE, et al. The diagnosis and management of

non-alcoholic fatty liver disease: practice guideline by the American Gastroenterological Association, American Association for the Study of Liver Diseases, and American College of Gastroenterology. Gastroenterology 2012;142:1592-609.

8. American College of Radiology. Liver imaging reporting & data system (LI-RADS®) (https://www.acr.org/Clinical-Resources/Clinical-Tools-and-Reference/Reporting-and-Data-Systems/LI-RADS).

9. Mazzaferro V, Regalia E, Doci R, et al. Liver transplantation for the treatment of small hepatocellular carcinomas in patients with cirrhosis. N Engl J Med 1996;334:693-9.

10. Stahl EP, Dhindsa DS, Lee SK, Sandesara PB, Chalasani NP, Sperling LS. Nonalcoholic fatty liver disease and the heart: JACC state-of-the-art review. J Am Coll Cardiol 2019;73:948-63.

11. Kasper P, Martin A, Lang S, et al. NAFLD and cardiovascular diseases: a clinical review. Clin Res Cardiol 2021;110:921-37.

12. Ismaiel A, Dumitraşcu DL. Cardiovascular risk in fatty liver disease: the liver-heart axis-literature review. Front Med (Lausanne) 2019;6:202.

13. Han E, Lee YH. Non-alcoholic fatty liver disease: the emerging burden in cardiometabolic and renal diseases. Diabetes Metab J 2017;41:430-7.

14. Salah HM, Pandey A, Soloveva A, et al. Relationship of nonalcoholic fatty liver disease and heart failure with preserved ejection fraction. JACC Basic Transl Sci 2021;6:918-32.

15. Ismaiel A, Colosi HA, Rusu F, Dumitraşcu DL. Cardiac arrhythmias and electrocardiogram modifications in non-alcoholic fatty liver disease. a systematic review. J Gastrointestin Liver Dis 2019;28:483-93.

16. Yalamanchili K, Saadeh S, Klintmalm GB, Jennings LW, Davis GL. Nonalcoholic fatty liver disease after liver transplantation for cryptogenic cirrhosis or nonalcoholic fatty liver disease. Liver Transpl 2010;16:431-9.

17. Dureja P, Mellinger J, Agni R, et al. NAFLD recurrence in liver transplant recipients. Transplantation 2011;91:684-9.